Question

$\mathbf{X}$ is a basic carbonate of metal $\mathbf{M}$, possibly containing water of crystallization. Two samples of equal mass were taken and treated separately: one was thoroughly treated in hydrogen, collecting $2.16\ \mathrm{g}$ of water, and the remaining residue was the element of metal $\mathbf{M}$, weighing $10.36\ \mathrm{g}$; the other was treated in oxygen, and after complete reaction, $0.448\ \mathrm{L}$ of carbon dioxide was obtained $(101.325\ \mathrm{kPa},\ 0°\mathrm{C})$, and the remaining residue was the highest oxide of $\mathbf{M}$. Then the metal $\mathbf{M}$ and the carbon:oxygen ratio in $\mathbf{X}$ are respectively:

A. All other options are incorrect.

B. Cu,1:5

C. Cu,1:8

D. Cu,4:15

E. Mn,1:5

F. Mn,1:8

G. Mn,4:15

H. Zn,1:5

I. Zn,1:8

J. Zn,4:15

K. Pb,1:5

L. Pb,1:8

M. Pb,4:15

N. U,1:5

O. U,1:8

P. U,4:15

Q. La,1:5

R. La,1:8

S. La,4:15

Answer

Correct Answer: L

Detailed Explanation

Assuming the chemical formula of $\mathbf{X}$ is $\mathbf{M}(\mathrm{OH})_\mathrm{x}(\mathrm{CO_3})_\mathrm{y} \cdot \mathrm{zH_2O}$,

CHECKPOINT 1 PTS

Set up unknowns to represent the chemical formula of $\mathbf{X}$ as $\mathbf{M}(\mathrm{OH})_\mathrm{x}(\mathrm{CO_3})_\mathrm{y} \cdot \mathrm{zH_2O}$

then the equations for the two reactions are respectively:

$$\mathbf{M}(\mathrm{OH})_\mathrm{x}(\mathrm{CO_3})_\mathrm{y} \cdot \mathrm{zH_2O} + (\tfrac{\mathrm{x}}{2} + \mathrm{y})\mathrm{H_2} \rightarrow \mathrm{yCO_2} + (\mathrm{x}+\mathrm{y}+\mathrm{z})\mathrm{H_2O} + \mathbf{M}$$

and

$$\mathbf{M}(\mathrm{OH})_\mathrm{x}(\mathrm{CO_3})_\mathrm{y} \cdot \mathrm{zH_2O} + (\tfrac{2\mathrm{a}-\mathrm{x}-2\mathrm{y}}{4})\mathrm{O_2} \rightarrow (\tfrac{\mathrm{x}}{2} + \mathrm{z})\mathrm{H_2O} + \mathrm{yCO_2} + \mathbf{M}O_a$$

Thus, we obtain the equation:

$$\frac{x+y+z}{y} = \frac{2.16/18.016}{0.448/22.4}$$

Simplifying, we get $x + z = 5y$

CHECKPOINT 1 PTS

Deduce the relationship between unknowns $x + z = 5y$

From the mass of the metal, we can list the second equation:

$$\frac{10.36}{\mathrm{M}} = \frac{2.16}{18.016(x+y+z)} = \frac{2.16}{18.016\cdot(6y)} = \frac{1}{50.04y}$$

Try to solve the equation. When $y = 2/5$, there is a reasonable solution Pb.

CHECKPOINT 1 PTS

Obtain $\mathbf{M}$ as Pb by solving the indeterminate equation

At this time, $x + z = 5y = 2$

The common valence states of Pb are +2 and +4. Here, it is obviously +2, so from the valence, we can calculate $x = \frac{6}{5}$, thus $z = \frac{4}{5}$

Therefore, $\mathbf{X}$ should be $\mathrm{Pb(OH)}_{\frac{6}{5}}(\mathrm{CO_3})_{\frac{2}{5}} \cdot \frac{4}{5}\mathrm{H_2O}$.

CHECKPOINT 1 PTS

The chemical formula of $\mathbf{X}$ is $\mathrm{Pb(OH)}_{\frac{6}{5}}(\mathrm{CO_3})_{\frac{2}{5}} \cdot \frac{4}{5}\mathrm{H_2O}$

Therefore, choose option L